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OS Lab 5 Tasks

1)

```
…/lab5/task1 main • ? ❯ make
gcc main.c book.c issue.c -o library

…/lab5/task1 main • ? ❯ ./library
Library System
1. Add Book
2. Issue Book
0. Exit
Enter: 1
Enter Book ID: 1
Enter Book Name: Reverend Insanity
Book Added!
Library System
1. Add Book
2. Issue Book
0. Exit
Enter: 2
Enter Book ID to issue: 2
Book Not Found!
Library System
1. Add Book
2. Issue Book
0. Exit
Enter: 2
Enter Book ID to issue: 1
Book Issued to Student 1!
Library System
1. Add Book
2. Issue Book
0. Exit
Enter: 1
Enter Book ID: 2
Enter Book Name: Harry Potter
Book Added!
Library System
1. Add Book
2. Issue Book
0. Exit
Enter: 2
Enter Book ID to issue: 2
Book Issued to Student 2!
Library System
1. Add Book
2. Issue Book
0. Exit
Enter: 0
Exiting...

…/lab5/task1 main • ? ❯ make clean
rm library

…/lab5/task1 main • ? ❯
```

// Makefile

.PHONY = clean

```
defmake:
	gcc main.c book.c issue.c -o library

clean:
	rm library

// library.h
#pragma once

#define MAX_BOOKS 100

typedef struct {
  int id;
  char name[100];
  int isIssued;
} Book;

extern Book library[MAX_BOOKS];
extern int bookCount;
void addBook();
void issueBook();

// book.c
#include "library.h"
#include <stdio.h>

Book library[MAX_BOOKS];
int bookCount;

void addBook() {
  if (bookCount >= MAX_BOOKS) {
    printf("Library is full.\n");
    return;
  }
```

```
  printf("Enter Book ID: ");
  scanf("%d", &library[bookCount].id);

  printf("Enter Book Name: ");
  scanf(" %[^\n]", library[bookCount].name);

  library[bookCount].isIssued = 0;
  bookCount++;

  printf("Book Added!\n");
}

// issue.c
#include "library.h"
#include <stdio.h>

void issueBook() {
  int id;
  printf("Enter Book ID to issue: ");
  scanf("%d", &id);

  for (int i = 0; i < bookCount; i++) {
    if (library[i].id == id) {
      if (library[i].isIssued) {
        printf("Book Already Issued!\n");
      } else {
        library[i].isIssued = 1;
        printf("Book Issued to %d!\n", i);
      }

      return;
    }
  }

  printf("Book Not Found!\n");
```

```
}

// main.c
#include "library.h"
#include <stdio.h>

int main() {
  int choice;

  while (1) {
    printf("Library System\n");
    printf("1. Add Book\n");
    printf("2. Issue Book\n");
    printf("0. Exit\n");

    printf("Enter: ");
    scanf("%d", &choice);

    switch (choice) {
    case 1:
      addBook();
      break;
    case 2:
      issueBook();
      break;
    case 0:
      printf("Exiting...\n");
      return 0;
    default:
      printf("Invalid Choice. Try Again.\n");
    }
  }

  return 1;
}
```

2)

```
…/lab5/task2 main ❯ mkdir Project_Audit

…/lab5/task2 main ? ❯ cd Project_Audit/

…/task2/Project_Audit main ? ❯ mkdir Reports Logs

…/task2/Project_Audit main ? ❯ ls
Permissions Size User   Date Modified Name
drwxr-xr-x     - f4ll3n  5 Mar 17:05  Logs
drwxr-xr-x     - f4ll3n  5 Mar 17:05  Reports

…/task2/Project_Audit main ? ❯ cd Reports/

…/Project_Audit/Reports main ? ❯ date
Thu Mar  5 05:06:41 PM PKT 2026

…/Project_Audit/Reports main ? ❯ touch report1.txt

…/Project_Audit/Reports main ? ❯ report1.txt < date
bash: date: No such file or directory

…/Project_Audit/Reports main ? ✗ date > report1.txt

…/Project_Audit/Reports main ? ❯ cat report1.txt
Thu Mar  5 05:10:20 PM PKT 2026

…/Project_Audit/Reports main ? ❯ nvim report1.txt
```

```
Thu Mar  5 05:10:20 PM PKT 2026
Lorem ipsum dolor sit amet consectetur adipiscing elit.-
Consectetur adipiscing elit quisque faucibus ex sapien vitae.-
Error sapien vitae pellentesque sem placerat in id.-
Placerat in error cursus mi pretium tellus duis.-
Pretium tellus duis convallis tempus leo eu aenean.
```

NORMAL main OS-Lab-Tasks report1.txt i 3 83% 5:17 17:18

```
1   Thu Mar  5 05:10:20 PM PKT 2026
2   Lorem ipsum dolor sit amet consectetur adipiscing elit.-
3   Consectetur adipiscing elit quisque faucibus ex sapien vitae.-
4   Ex sapien vitae pellentesque sem placerat in id.-
```

NORMAL main OS-Lab-Tasks report2.txt ^[3 Bot 4:1 17:17

```
…/Project_Audit/Reports main ? ❯ nvim report1.txt

…/Project_Audit/Reports main ? ❯ cp report1.txt report2.txt

…/Project_Audit/Reports main ? ❯ nvim re
...1.txt  ...2.txt

…/Project_Audit/Reports main ? ❯ nvim report2.txt

…/Project_Audit/Reports main ? ❯ nvim report2.txt

…/Project_Audit/Reports main ? ❯ nvim repo
...1.txt  ...2.txt

…/Project_Audit/Reports main ? ❯ nvim report1.txt

…/Project_Audit/Reports main ? ❯ grep -i "error" report1.txt summary.txt
report1.txt:Error sapien vitae pellentesque sem placerat in id.
report1.txt:Placerat in error cursus mi pretium tellus duis.
grep: summary.txt: No such file or directory

…/Project_Audit/Reports main ? ✗ touch summary.txt

…/Project_Audit/Reports main ? ❯ grep -i "error" report1.txt summary.txt
report1.txt:Error sapien vitae pellentesque sem placerat in id.
report1.txt:Placerat in error cursus mi pretium tellus duis.

…/Project_Audit/Reports main ? ❯ wc -lw report1.txt report2.txt
  6  47 report1.txt
  4  31 report2.txt
 10  78 total

…/Project_Audit/Reports main ? ❯ # lines words filename

…/Project_Audit/Reports main ? ❯ ls -l
Permissions Size User   Date Modified Name
.rw-r--r--   307 f4ll3n  5 Mar 17:18  report1.txt
.rw-r--r--   202 f4ll3n  5 Mar 17:17  report2.txt
.rw-r--r--     0 f4ll3n  5 Mar 17:21  summary.txt

…/Project_Audit/Reports main ? ❯ cat summary.txt

…/Project_Audit/Reports main ? ❯ grep -i "error" report1.txt > summary.txt

…/Project_Audit/Reports main ? ❯ cat summary.txt
Error sapien vitae pellentesque sem placerat in id.
Placerat in error cursus mi pretium tellus duis.

…/Project_Audit/Reports main ? ❯ chmod u+rwx summary.txt

…/Project_Audit/Reports main ? ❯ ls -l
Permissions Size User   Date Modified Name
.rw-r--r--   307 f4ll3n  5 Mar 17:18  report1.txt
.rw-r--r--   202 f4ll3n  5 Mar 17:17  report2.txt
.rwxr--r--   103 f4ll3n  5 Mar 17:28  summary.txt

…/Project_Audit/Reports main ? ❯
```

```
  6  47 report1.txt
  4  31 report2.txt
 10  78 total

…/Project_Audit/Reports main ? ❯ # lines words filename

…/Project_Audit/Reports main ? ❯ ls -l
Permissions Size User   Date Modified Name
.rw-r--r--   307 f4ll3n  5 Mar 17:18  report1.txt
.rw-r--r--   202 f4ll3n  5 Mar 17:17  report2.txt
.rw-r--r--     0 f4ll3n  5 Mar 17:21  summary.txt

…/Project_Audit/Reports main ? ❯ cat summary.txt

…/Project_Audit/Reports main ? ❯ grep -i "error" report1.txt > summary.txt

…/Project_Audit/Reports main ? ❯ cat summary.txt
Error sapien vitae pellentesque sem placerat in id.
Placerat in error cursus mi pretium tellus duis.

…/Project_Audit/Reports main ? ❯ chmod u+rwx summary.txt

…/Project_Audit/Reports main ? ❯ ls -l
Permissions Size User   Date Modified Name
.rw-r--r--   307 f4ll3n  5 Mar 17:18  report1.txt
.rw-r--r--   202 f4ll3n  5 Mar 17:17  report2.txt
.rwxr--r--   103 f4ll3n  5 Mar 17:28  summary.txt

…/Project_Audit/Reports main ? ❯ pwd
/home/f4ll3n/Work/OS/OS-Lab-Tasks/lab5/task2/Project_Audit/Reports

…/Project_Audit/Reports main ? ❯ date
Thu Mar  5 05:30:28 PM PKT 2026

…/Project_Audit/Reports main ? ❯ df -h
Filesystem      Size  Used Avail Use% Mounted on
dev             7.6G     0  7.6G   0% /dev
run             7.7G  1.7M  7.7G   1% /run
efivarfs        374K  259K  111K  71% /sys/firmware/efi/efivars
/dev/nvme0n1p8  110G   47G   59G  45% /
tmpfs           7.7G  142M  7.6G   2% /dev/shm
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M     0  1.0M   0% /run/credentials/systemd-networkd.service
tmpfs           7.7G   90M  7.6G   2% /tmp
/dev/nvme0n1p7 1022M  735M  288M  72% /boot/efi
tmpfs           1.6G  1.8M  1.6G   1% /run/user/1000

…/Project_Audit/Reports main ? ❯ du -sh ~/
12G     /home/f4ll3n/

…/Project_Audit/Reports main ? ❯ ls
Permissions Size User   Date Modified Name
.rw-r--r--   307 f4ll3n  5 Mar 17:18  report1.txt
.rw-r--r--   202 f4ll3n  5 Mar 17:17  report2.txt
.rwxr--r--   103 f4ll3n  5 Mar 17:28  summary.txt

…/Project_Audit/Reports main ? ❯
```

3)

```
arch-host

…/lab5/task3 main ? ❯ chmod u+x result.sh

…/lab5/task3 main ? ❯ ./result.sh
Enter student's name: Aly
Enter student's marks in three subjects ( like m1 m2 m3 ): 60 70 80
Student Aly's Grade: B
Average: 70
…/lab5/task3 main ? ❯ ./result.sh
Enter student's name: test
Enter student's marks in three subjects ( like m1 m2 m3 ): 80 23 45
Student test's Grade: F
Average: 49
…/lab5/task3 main ? ❯
```

```
// result.sh
#!/bin/bash

printf "Enter student's name: "
read name

printf "Enter student's marks in three subjects ( like m1 m2 m3 ): "
read mark1 mark2 mark3

sum=$((mark1 + mark2 + mark3))

average=$((sum / 3))

if [[ $average -ge 80 ]]; then
  result='A'
elif [[ $average -ge 70 ]]; then
  result='B'
elif [[ $average -ge 60 ]]; then
  result='C'
elif [[ $average -ge 50 ]]; then
  result='D'
else
```

```
  result='F'
fi

printf "Student %s's Grade: %c \nAverage: %d" $name $result $average
```

4)

```
…/lab5/task4 main • ? ❯ ./analyze.sh
Script Name: ./analyze.sh
Total Args: 0
No Arguments passed.

Current User: f4ll3n
Home Directory:
Desktop
Documents
Downloads
Music
Pictures
Public
Templates
Videos
Wallpapers
Work

Current Directory: /home/f4ll3n/Work/OS/OS-Lab-Tasks/lab5/task4
/home/f4ll3n/Work/OS/OS-Lab-Tasks/lab5/task4/bin
/home/f4ll3n/.local/share/mise/installs/node/25.2.0/bin
/home/f4ll3n/.local/share/mise/installs/python/3.14.2/bin
/home/f4ll3n/.local/share/omarchy/bin
/home/f4ll3n/.local/share/omarchy/bin
/home/f4ll3n/.local/share/../bin
/usr/local/sbin
/usr/local/bin
/usr/bin
/usr/lib/jvm/default/bin
/usr/bin/site_perl
/usr/bin/vendor_perl
/usr/bin/core_perl
/home/f4ll3n/.local/bin
```

```
…/lab5/task4 main • ? ❯ ./analyze.sh 2 3 "Hello"
Script Name: ./analyze.sh
Total Args: 3
Arg 1: 2
Arg 2: 3
Arg 3: Hello

First Arg: 2
Second Arg: 3

Current User: f4ll3n
Home Directory:
Desktop
Documents
Downloads
Music
Pictures
Public
Templates
Videos
Wallpapers
Work

Current Directory: /home/f4ll3n/Work/OS/OS-Lab-Tasks/lab5/task4
/home/f4ll3n/Work/OS/OS-Lab-Tasks/lab5/task4/bin
/home/f4ll3n/.local/share/mise/installs/node/25.2.0/bin
/home/f4ll3n/.local/share/mise/installs/python/3.14.2/bin
/home/f4ll3n/.local/share/omarchy/bin
/home/f4ll3n/.local/share/omarchy/bin
/home/f4ll3n/.local/share/../bin
/usr/local/sbin
/usr/local/bin
/usr/bin
/usr/lib/jvm/default/bin
/usr/bin/site_perl
/usr/bin/vendor_perl
/usr/bin/core_perl
/home/f4ll3n/.local/bin

…/lab5/task4 main • ? ❯
```

```
// analyze.sh
#!/bin/bash

printf "Script Name: %s\nTotal Args: %d\n" "$0" "$#"

i=0
for arg in "$@"; do
```

```
  i=$((i + 1))
  printf "Arg %d: %s\n" "$i" "$arg"
done


if [ -n "$1" ]; then
  printf "\n"
  printf "First Arg: %s\n" "$1"
fi

if [ -n "$2" ]; then
  printf "Second Arg: %s\n" "$2"
fi

if [ "$#" -eq 0 ]; then
  printf "No Arguments passed.\n"
fi

printf "\n"
printf "Current User: %s\n" "$(whoami)"
printf "Home Directory:\n%s\n\n" "$(ls ~/)"

printf "Current Directory: %s\n" "$(pwd)"

printf '%s\n' "${PATH//:/$'\n'}"
```